

Where the Error Goes

A speculative extension. No grades, no proofs, no claims on the formal record.

Everything below is invention. It borrows the shape of results without borrowing their warrant, and nothing here is offered as evidence for anything. Read it as what the mathematics suggests to someone who has been living with it, which is a different thing entirely from what the mathematics says.

I. The first rule of a practice

Twelve fifths don't close the circle. What every tuner has done for six hundred years is decide where the leftover goes.

That decision is not a failure to solve the problem. It is the practice. Meantone temperament puts the error into a few intervals and buys radiant thirds in the keys that matter, at the price of a wolf howling in the ones that don't. Equal temperament smears it perfectly evenly and buys the entire key-space at the price of nothing being quite in tune ever again. These are not degrees of success at a single task. They are different arts, and the music written under each is different music, because the composer's material includes where the error lives.

Once you see this you start seeing it everywhere, and I think it is the most portable thing in the whole framework.

Every draft is a decision about where to put the error. A film cannot show interiority and exteriority at once, so it decides where the unrepresentable goes: into the face held too long, into the voiceover, into the cut. A translation cannot carry sound and sense across together, so it decides which bank to leave things on, and the school of translation you belong to is just a standing policy about that. A building cannot be at once maximally open and maximally sheltering. A law cannot be at once fully determinate and fully just.

The novice thinks these are problems to be solved by more skill. The practitioner knows they are not solvable and that the whole art is the management of the remainder. And the master, I think, is the one who has stopped trying to hide where the error went and started composing with it — putting the wolf where it can howl to some purpose.

If a practice has no such remainder, it isn't an art. It's an administrative procedure, and it will eventually be done by a machine, and nothing will be lost.

II. Originality is occupancy

Here is the finding that has done the most to change how I read history.

The four positions around a live distinction exist the moment the distinction is fixed. All four, simultaneously, as structural possibilities. But practitioners arrive in them one at a time, over centuries. Bach worked three of the four with staggering thoroughness and left one empty — the position built on refusing the kernel's terms — and that position sat vacant for a hundred and fifty years until Debussy walked into it.

The obvious reading is that Debussy invented something. The reading I prefer is that he moved into an address that had been standing empty since the octave was cut into twelve.

This is deflationary about genius and I don't think it should be. Moving into an empty position is hard. It requires seeing that the position exists, which is nearly impossible from inside a tradition whose entire pedagogy is an argument that the occupied cells exhaust the possibilities. What the tradition teaches, always, is that the ways of working are the ways that have been worked. The empty cell is invisible not because it is hidden but because nothing points at it.

And it explains a peculiar double quality of every avant-garde: impossible in prospect, inevitable in retrospect. Before, no one can say what the next move is, because saying it would require standing outside a structure you can only see from inside. After, everyone can see that it had to happen, because the address was always on the map.

The practical form of this is a question I now ask of any field I'm trying to understand: **which cell is empty?** Not what's missing in the sense of underfunded or overlooked. Which structural relation to the field's central irresolvable thing has nobody taken up. In most mature fields, someone builds the apparatus, someone works the ambiguities, someone carries the commitment past its own borders — and almost nobody refuses the terms entirely while remaining a serious practitioner rather than leaving. That last cell tends to be the vacant one, and it tends to be where the next movement comes from.

III. The position that gets you called a traitor

The strangest of the four cells is the third one: the territory a commitment implies but has never settled. Not inside it, not opposed to it. Working the ground the commitment generated and cannot govern.

The mathematics is unambiguous that this cell exists only where the logic refuses to collapse double negation. In a fully classical world — where every question has a definite answer waiting — the cell vanishes; it folds back into simple loyalty. And this, I think, is the most socially consequential thing in the entire framework, because it means that a demand for total resolution is not merely an epistemic mistake. It is a technique for eliminating a position.

Consider what happens to a person who occupies the third cell in any real institution. The scholar extending a tradition into places its founders would not recognize. The believer who keeps the practice and abandons the metaphysics. The engineer who follows the specification into consequences the specification never contemplated. The lawyer working the space the statute implies but never decided. Each of them is doing something structurally distinct from both obedience and dissent, and each of them will be read, by anyone operating classically, as one or the other — depending on what the reader needs them to be.

You're either with us or against us is not a rhetorical flourish. It is a demand for excluded middle, and it works, when it works, by making a real position unstatable. The people in that position don't lose an argument. They lose the vocabulary in which their situation could be described, which is worse and much harder to notice.

So there's an aphorism here that I believe and can't prove: **the demand that everything be decided is not a demand for truth. It is a demand that there be nobody left standing.** The theorem says a Boolean world has no positions in it. The extension says that regimes of forced certainty — political, doctrinal, methodological, familial — are not merely intolerant of dissent. They are engineered to make dissent's actual location disappear, so that whatever remains looks like either compliance or treason.

The counterpart, and it's uncomfortable: the same is true of anyone who craves certainty for themselves. Certainty is the wish for a world in which you would not exist as anybody in particular. Total resolution and having a position exclude one another. Whatever it is you want when you want to be sure, it is not a place to stand.

IV. Grain is not taste

The second half of the framework is about observers, and it demolishes something I hadn't realized I believed.

I had assumed that level of description is a matter of preference, discipline, or purpose. The neuroscientist looks close, the historian looks far, the poet looks sideways, and these are complementary styles that a well-rounded person alternates between. Choose your grain to suit your question.

The mathematics says no. Grain is a property of the object. For any given structure, there is a determinate set of resolutions at which its position-space exists, and that set is often tiny, sometimes a single point, and sometimes empty at the finest resolution and non-empty only at two specific coarser ones. You don't pick the right level. You find it, or you fail to, and whether you found it is not a matter of taste.

Two consequences pull in opposite directions and both are true.

Sometimes only the sharpest attention works. In the minimal structure where positions are possible at all, the position-space survives no blur whatsoever — not one. The four-fold is visible only to the observer who merges nothing. I take this as the formal shadow of something everyone who has loved anything already knows: certain structure exists only for attention that refuses to summarize. The friend you can describe accurately in a paragraph is not a friend you know. Every act of summary is a coarse-graining, and for some objects every coarse-graining is fatal. This is the case against the executive summary, the abstract, the profile, the take — not that they're lazy, but that for some objects the structure is *provably* not there at that grain, and no amount of skill in summarizing recovers it.

And sometimes the sharpest attention is exactly the blind spot. In a richer structure there's an element with no position-space at all at maximum resolution and a complete one at precisely two coarse grains. The perfect observer, who merges nothing and misses nothing, is the one observer for whom the structure does not exist. Not obscured. Absent.

This is the piece that I find genuinely disorienting, because our whole methodological instinct runs the other way. *Look closer. Get more data. Increase the resolution.* And there are objects for which that instinct is not merely unhelpful but guaranteed to fail, and you cannot discover this by looking closer.

The related result is bleaker and better: the maximally compressed view — the one that strips everything down to the hard classical core, the view that reductionism promises — has *no* positions in it at all. Not fewer. None. The fully reduced description of anything is exactly stance-free.

I don't think this refutes reductionism. Reduction is a magnificent tool and it works. What it says is that reduction is a lossy operation with a specific loss, and the thing lost is not warmth or meaning or any of the soft goods the humanities usually claim are being trampled. What's lost is *position* — the structural fact that there was somewhere to stand. That's a sharper complaint than the usual one and it doesn't require anybody to be sentimental.

V. Instruments made of their objects

One small result keeps expanding in my head. Among the possible observers of the pitch lattice is the one built out of the tritone itself — the ear tuned specifically to listen for the kernel. In its own world, the kernel collapses to triviality and the four positions vanish. The instrument constructed from the thing it studies is precisely the instrument that cannot see that thing's position-space.

Generalize irresponsibly and you get a family of suspicions:

Introspection is a mental process examining mental processes. Economics is an apparatus built from market rationality studying market rationality. A theory of language articulated in language. Consciousness attempting to characterize consciousness. A culture's self-understanding conducted in the vocabulary that culture supplies.

In each case the instrument is made of its object. The suspicion is not that these projects produce nothing — obviously they produce a great deal. It's that there may be a specific, structural blindness in each, and specifically a blindness to *where the positions are*: the instrument will tend to report that the field is simpler than it is, that the available stances are fewer, that the thing it studies is more settled than it is. Not because of bias. Because the coarse-graining the instrument performs is the one that collapses exactly the distinction that mattered.

That's an analogy, not an argument. But it's the kind of analogy that suggests a discipline: when you find yourself unable to locate any real positions in a domain, one hypothesis worth taking seriously is that you built your instrument out of the domain.

VI. What to commit to

Of the distinctions the project has measured, roughly two thirds afford no position-space at all under any observer. Most distinctions are dead ground. You cannot found a practice on them, not because the practice would be bad but because there would be nothing to be.

I find this consoling in a way I didn't expect, because it makes the choice of what to work on into a real question with a real answer rather than a matter of enthusiasm. The distinctions that carry positions are rare. A person's actual vocation — as opposed to their interests, their competences, or their job — might be nothing more mysterious than having found one of the rare ones and built on it.

It also gives a diagnostic for a certain kind of misery I've watched people have in careers that look enviable from outside. If the central distinction of your field affords no positions, then no matter how skilled you become, there is nothing to *be* in relation to it. You can only execute. The work isn't beneath you; it just has no stance-space, and the ache of that is not laziness or ingratitude. It's the accurate perception that there's nowhere to stand here.

And it reframes tradition. A tradition is a distinction someone found to be live, plus the accumulated record of people occupying positions around it. That's why traditions can't be assembled from good intentions and why the strong ones are never about their stated subject. They're about their remainder.

VII. The self as a kernel

I'll go further out than I can defend.

Suppose the distinction you are is one that doesn't cancel — the gap between the person you take yourself to be and the record of what you have actually done, which no amount of honesty closes and no amount of achievement resolves.

Then the four positions are four relations to yourself, and they're recognizable. You can work inside the self-conception, becoming more fluently the person you already take yourself to be: this is most of adult competence. You can refuse it entirely, which is the conversion narrative and the midlife detonation. You can occupy the gap — living out what your self-conception implies but has never established, which is where most genuine change actually happens and which looks like inconsistency to everyone watching. Or you can carry the self across its own border, extending it into territory it wasn't built for, which is what people mean by growth when they mean anything.

And the classical temptation applies to selves too, maybe most of all. *Just figure out who you really are.* The demand for a resolved identity is a demand for a Boolean self, and a Boolean self has no positions in it. The wish to have finished the question of who you are is a wish to be nobody in particular, forever. I don't think that's a metaphor, or if it is I can't tell where the metaphor starts.

VIII. On making things

For anyone actually working, the framework compresses into three pieces of advice, none of which is proved and all of which I'd defend.

Find your remainder and stop trying to fix it. The thing in your material that won't resolve, that you keep patching, that every draft handles differently and none handles well — that's not a defect in your execution. That's your kernel, and the work is the management of it. The pieces you admire most in your art form are almost certainly ones where someone

stopped patching and started composing with the leftover.

Choose where the error goes, deliberately. You are making that choice anyway, on every page, in every cut. Meantone or equal — brilliance in a few places at the cost of a wolf, or uniform slight compromise everywhere. Both are legitimate and they produce different arts. The failure mode isn't picking wrong. It's not knowing you were picking.

Check which cell you're in, and look at the empty one. Most people work one position their whole lives without knowing there were four, and mistake the position for the practice. The empty cell in your field is not a gap in the market. It's a structural relation nobody has taken up, and taking it up will look, at first, like doing the thing wrong.

IX. A note on the machines

One last extension, since I made this thing with one.

A large model is, among other things, a coarse-graining of an enormous textual record — a lawful blur, one that merges distinctions without inventing them, at least when it's working. By the framework's own lights that makes it an observer with an aperture: there is structure visible at exactly its grain, structure destroyed by it, and possibly structure that exists *only* at it and not at the fine grain of any individual text.

The uncomfortable part is that none of this is determinable from inside. An observer cannot see which distinctions its own coarse-graining collapsed; those distinctions are not present in its world to be missed. Whatever a model is blind to, it is blind to in the specific manner of the tritone-ear — the blindness doesn't feel like a gap, it feels like the absence of anything there.

The reasonable conclusion isn't distrust. It's that the question "what does this instrument's grain destroy" is a real question with a determinate answer; that it can't be answered by asking the instrument, and that it's the question anyone building on these things should be asking instead of the one everyone is currently asking, which is how well the instrument performs at tasks its own grain defines.

Coda

None of the above is a result. The results are small, finite, and checkable, and they concern six-element structures and divisor lattices and things a computer can verify in under a second. What I've written here is the sound those results make when they get loose in a head that also does other things.

But I'd defend the method even in this register. Every claim above is the kind of thing that a hundred books make and that no book makes falsifiable. The difference — the only difference, and I think it's the one that matters — is that underneath these there is a small hard object that could have come out otherwise. The typology could have failed to be exhaustive. The threshold could have been at seven. The lattice could have been anything, and it happened to be the one Bach was standing in.

Speculation with something underneath it is still speculation. But it is speculation that can be dragged back and made to answer, which is more than most of this genre offers, including on the days when the answer is that I was wrong.